

# Data Science Capstone Project Report

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## SpaceX Falcon 9 First-Stage Landing Prediction

Author: [Your name] | Date: [Date]

GitHub: <https://github.com/YOUR-USERNAME/YOUR-REPO>

# Executive Summary

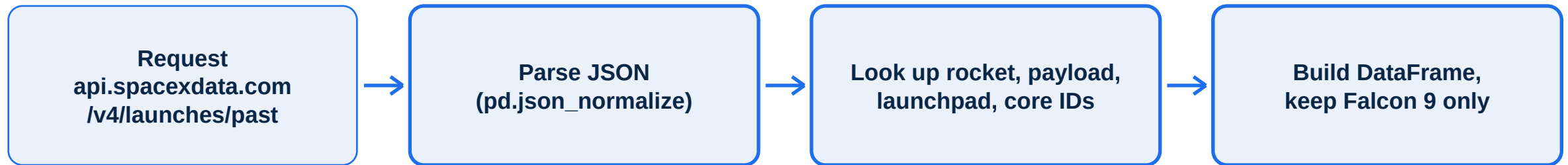
- **Methods:** data collection (SpaceX REST API + Wikipedia web scraping), data wrangling, EDA with visualization and SQL, interactive analytics (Folium, Plotly Dash), predictive modeling.
- **Models:** Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV.
- **Key results:** landing success rate rises with flight number and over time; orbits such as ES-L1, GEO, HEO, SSO show the highest success; KSC LC-39A has the best site success rate.
- **Best model:** [Model name] with test accuracy [xx.x%] - reliably predicts first-stage landing outcome.

# Introduction

- **Background:** SpaceX advertises Falcon 9 launches at about \$62 million, versus \$165 million+ for other providers - the saving comes from reusing the first stage.
- If we can predict whether the first stage will land, we can estimate the cost of a launch.
- **Problem:** Given launch features (payload, orbit, launch site, booster version, etc.), predict whether the Falcon 9 first stage lands successfully.
- **Value:** useful for a competitor bidding against SpaceX for a rocket launch.

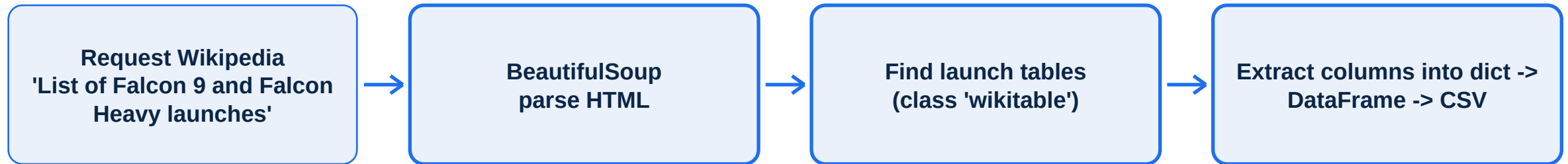
# Data Collection - SpaceX API

- Key phrases: **requests.get()**, **.json()**, helper functions `getBoosterVersion` / `getLaunchSite` / `getPayloadData` / `getCoreData`

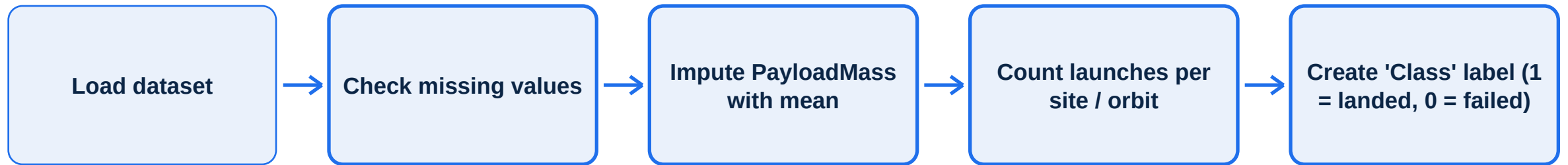


# Data Collection - Web Scraping

- Key phrases: `requests.get()`, `BeautifulSoup(html.parser)`, `find_all('table')`, `pd.DataFrame(launch_dict)`



# Data Wrangling Methodology



- Outcomes such as True ASDS, True RTLS, True Ocean = **1**; False ASDS, False RTLS, None None = **0**.
- Result: cleaned dataset exported to CSV for EDA and modeling.

# EDA with Data Visualization

- **Scatter plots:** Flight Number vs Launch Site, Payload vs Launch Site, Flight Number vs Orbit, Payload vs Orbit - reveal relationships between features and outcome.
- **Bar chart:** success rate by orbit - shows which orbits land most reliably.
- **Line chart:** yearly launch success trend - shows improvement over time.
- **Feature engineering:** one-hot encoding of Orbit, LaunchSite, LandingPad, Serial for modeling.

# EDA with SQL

- Loaded the SpaceX table into SQLite / Db2 and queried with SQL magic.
- Unique launch sites and sites beginning with 'CCA'.
- Total and average payload mass (NASA CRS, booster F9 v1.1).
- First successful ground-pad landing date.
- Boosters with drone-ship success and payload 4,000-6,000 kg.
- Mission outcome counts, max-payload boosters, 2015 failed drone-ship landings, and outcome ranking by date range.

# Interactive Visual Analytics

- **Folium map:** circles and markers for each launch site, green/red markers for success/failure using MarkerCluster, and lines with distances to coastline, railway, highway and city.
- **Plotly Dash app:** launch-site dropdown, payload range slider, success pie chart, and payload vs outcome scatter plot colored by booster version.

*Dash:* [https://github.com/YOUR-USERNAME/YOUR-REPO/blob/main/spacex\\_dash\\_app.py](https://github.com/YOUR-USERNAME/YOUR-REPO/blob/main/spacex_dash_app.py)

*Folium:* [https://github.com/YOUR-USERNAME/YOUR-REPO/blob/main/folium\\_map.ipynb](https://github.com/YOUR-USERNAME/YOUR-REPO/blob/main/folium_map.ipynb)

# EDA with Visualization - Results

- Flight Number vs Launch Site: later flights succeed more, especially at CCAFS SLC 40 and KSC LC 39A.
- Payload vs Launch Site: VAFB SLC 4E has no heavy-payload (>10,000 kg) launches.
- Success rate by orbit: ES-L1, GEO, HEO, SSO are ~100%; GTO is lowest.
- Yearly trend: success rate rises steadily from 2013 to 2020.

*[Insert scatter plots, orbit bar chart and yearly trend line chart]*

# EDA with SQL - Results

| Task                                     | Result                                             |
|------------------------------------------|----------------------------------------------------|
| Unique launch sites                      | CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E |
| Total payload (NASA CRS)                 | [value] kg                                         |
| Avg payload (F9 v1.1)                    | [value] kg                                         |
| First successful ground landing          | [date]                                             |
| Drone-ship success, 4-6 t payload        | [booster list]                                     |
| Mission outcomes                         | Success [n] / Failure [n]                          |
| Max-payload boosters                     | F9 B5 ... (list)                                   |
| 2015 failed drone-ship landings          | [month, booster, site]                             |
| Outcome ranking 2010-06-04 to 2017-03-20 | [ranked outcomes]                                  |

# Folium Map - Results

- Launch site markers: all sites near the coast (Florida and California).
- Launch records: green = success, red = failure, shown as clusters per site.
- Proximity: sites are close to coastline, railway and highway (~1 km) but far from cities - safety and logistics.

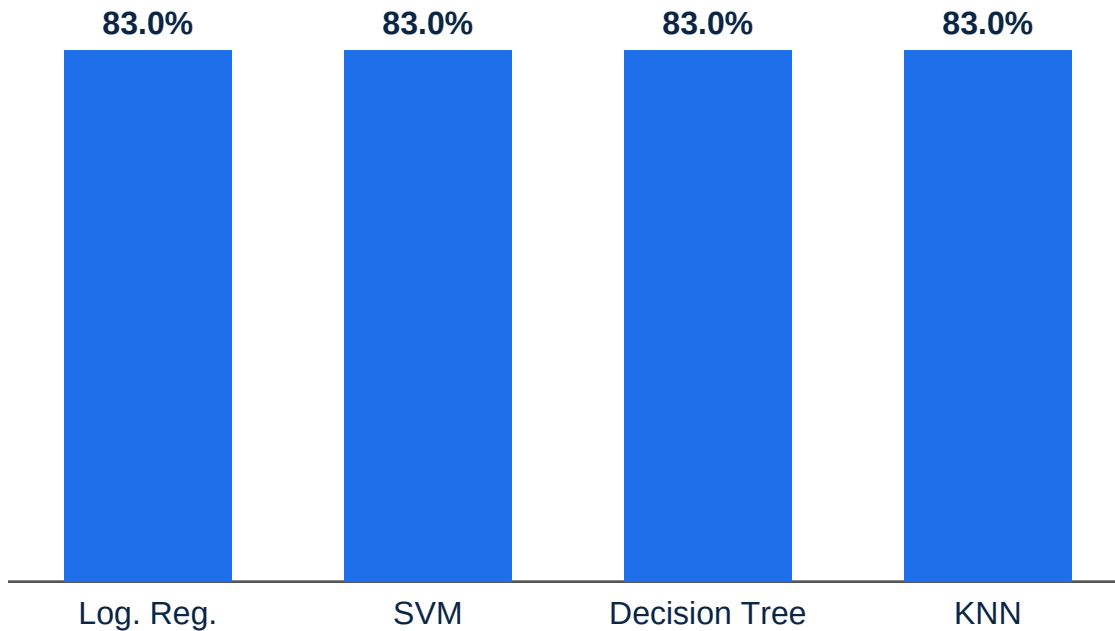
*[Insert Folium map screenshots: site markers, colored launch records, distance lines]*

# Plotly Dash - Results

- Pie chart (all sites): KSC LC-39A has the largest share of successful launches.
- Pie chart (best site): KSC LC-39A success vs failure ratio [xx% / xx%].
- Scatter plot: payload vs outcome by booster version - low/mid payloads (about 2,000-5,500 kg) show the highest success; FT booster performs best.

*[Insert Dash screenshots: success pie charts and payload vs outcome scatter plots]*

# Predictive Analysis



*Placeholder accuracy - replace with your GridSearchCV results*

## Confusion matrix (best model)

|                      | Pred: did not land | Pred: landed |
|----------------------|--------------------|--------------|
| Actual: did not land | [TN]               | [FP]         |
| Actual: landed       | [FN]               | [TP]         |

- **Best model:** [name] - highest cross-validation and test accuracy. Main error: false positives (predicts landing that fails).

# Conclusion

- Landing success improves with experience: later flights and newer boosters (FT / Block 5) land more reliably.
- Orbit, payload mass and launch site are the strongest predictors of success.
- Best classifier reaches [xx%] accuracy; a small test set means results should be validated with more data.
- **Insight:** the learning curve of reusability lets a competitor estimate SpaceX's launch cost and price bids accordingly.
- **Next steps:** add weather data, use ensemble models (Random Forest, XGBoost), deploy the Dash app.